

# Optimizing Resource Allocation and Congestion Management in MANETs: A Deep Reinforcement Learning Approach for Scalable and Secure Network Operations

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**Abstract**—In this paper, we propose a new DRL-based method for the purpose of network optimization, where a deep Q-learning algorithm is utilized as model-free approach for handling the decision-making problem in non-stationary environment. The approach follows state space mapping based on key performance indicators (KPIs): throughput, latency, energy consumption, and scalability. The agent learns optimal policies via interacting with environment in a reward-driven manner, with a epsilon-greedy policy and experience replay to achieve stable convergence. The model was thoroughly compared with classical rule-based, heuristic, and central algorithms. We find that the DRL model clearly outperforms traditional-based models in all performance metrics, with traditional models either performing poorly or requiring idealistic assumptions such as full network observability. The DRL algorithm not only has stable and convergent performance in QoS under diverse scenarios, but also has acceptable computational overhead, which verifies the feasibility of applying it to resource-limited devices. In general, the proposed DRL scheme is generic, flexible and efficient for real-time network optimization, and can be in conjunction with the existing systems as one of the intelligent agents.

**Keywords**—Deep Reinforcement Learning, Network Optimization, Quality of Service, Throughput, Latency, Resource-Constrained Nodes

## I. INTRODUCTION

Mobile Ad hoc Networks MANETs have become the most important enabler for many decentralized and self-organizing applications. They are vital in situations such as military operations, disaster relief, vehicular networks, or IoT-based smart environments, where a central infrastructure is unavailable or impractical. Unlike wired or wireless networks, MANETs operate on the basis of dynamic topology and self-reliant collaboration among nodes. MANETs are dynamic yet complex because, each node is not just a terminal but also a router. But this vast, distributed, and dynamic nature creates many challenges — especially in resource allocation and congestion management. One issue is to efficiently use limited resources like bandwidth, power,

and processing ability, as well as to ensure continuous data flow under high-latency scenarios. vanishes accordingly, is extremely helpful to maintain the performance and scalability of MANETs [1, 2]. Resource allocation in MANETs is a complex problem for the inherent constraints of the network. Nodes in a MANET usually run on batteries, restricting their power budget, and communicate via shared wireless mediums, which are vulnerable to interference and congestion. Dynamic resource allocation for shared resources, such as channel bandwidth, time slots, and energy, is essential to keep the network running over time [3]. Yet the lack of centralized control makes this even more difficult because decisions need to be made in a decentralized manner. On top of that, the frequent changes to network topology due to the mobility of nodes and constantly forming and breaking communication links results in a networking scenario that can become difficult to navigate [4]. The event demand appears to have caused a slight distortion in resource allocation mechanisms that must be resilient and emergent to these changing conditions to prevent bottlenecks and guarantee the overall performance of the network [5].

An additional challenge of MANETs is congestion management it has a direct effect on QoS and the reliability of data. In a MANET, congestion arises when there is an overwhelming demand for the network resources, causing packet loss, increased delay and lower throughput. Again, this becomes an issue in high-density situations like disaster relief or defence operations when many nodes may simultaneously try and communicate and compete for shared channels [6]. Conventional congestion control methods that rely on a static network with centralized authority do not match the features of MANET. Consequently, there is a surge in demand for innovative approaches capable of observing, alleviating [7], and averting congestion, while accommodating the decentralized and mobile characteristics of the network [8].

Recent progress in AI (artificial intelligence) and ML (machine learning) are opening new paths to handle the challenges related to the complexities of MANETs. Out of these, deep reinforcement learning or DRL [1, 2, 3] has

become a field of growing interest owing to its capacity of learning policies and optimizing them in high-dimensional, stochastic environments. The decision-making aptitude of reinforcement learning and the function approximation ability of deep neural networks has made deep reinforcement learning (DRL) a powerful tool to solve resource allocation and congestion management problems arising in MOBILE AD HOC Networks (MANETs) [8]. The dynamic nature of DRL enables it to adaptively change actions (e.g. channel allocation, routing, prioritization, etc.) based on the current state of the system by continuously monitoring the network performance over time. Moreover, the independent learning of DRL enables it to adapt to various types of network conditions and topology changes, which is the primary requirement in a MANET [9].

Despite that, there are some challenges using DRL for MANETs. This can be demanding for portable mobile devices running DRL algorithms in resource-constrained environments and recommends more data and computing resources than what is available in MANET nodes. In addition, ensuring the security and reliability of DRL-driven decisions is crucial in scenarios such as in the MANETs operating in the hostile environments. For example, attacks such as spoofing, jamming, and malicious routing can disrupt or exploit vulnerabilities in the learning process or the learned policies, which can lead to poor outcomes of the learned policies, potentially with catastrophic effects [10]. A holistic approach must be sustained to better integrate DRL and a novel security mechanism into the network availability with the vigilance over any malicious activity [11].

The way that DRL can change the scope of routing in resource allocation and congestion management in MANETs cannot be denied despite these challenges. Unlike classical rule-based approaches or heuristic methods, DRL with neural networks can learn optimal policies through direct interaction with the environment, thus performing without needing manual tuning or preset strategies [12]. In Dynamic networks such as MANET, due to the constantly changing topology of the network, static methods are insufficient. Moreover, the capability of DRL to capture complex, nonlinear dynamics renders it highly appropriate for dealing with the interdependencies inherent between resource allocation and congestion mitigation, facilitating joint optimization of network performance [13].

One of the major advantages of DRL is its scalability. Less traditional methodologies are ensuring performance are needed when the size and complexity of MANETs increase since traditional methodologies must rely on centralized control or static coverage solutions. DRL, in contrast to this requires very little processing and can be distributed throughout a network, with each node deciding locally based on lightweight models, but working together with others for optimization of their model globally [14]. However, this decentralized nature is not just in terms of ownership; it means that there is no one organization that has sole control, which reduces the overhead on individual nodes and makes it more resistant to failure, nor can the things in a blockchain be altered or removed after they are added, making it extremely secure against attacks. Additionally, the real-time operation of DRL [15] enables it to adapt rapidly to network changes, thereby ensuring robust QoS in dynamic environments.

Constructing frameworks based on DRL with suitable performance, scalability, and security in MANETs will enable us to capitalize on the potential of DRL, highlighting the need for use-case-specific adaptations. This includes creating efficient training algorithms adaptable to the limitations of MANET nodes, designing lightweight DRL models to reduce computational overhead, and ensuring the integration of adversarial attack detection and mitigation mechanisms, etc. Moreover, deploying DRL in the context of MANETs necessitates extensive testing and validation in realistic environments to ensure its practical utility and performance [16]. Therefore, simulation tools with testbed experimentation and actual deployments are vital in evaluating the performance of DRL-based methods and assist to identify areas that need focusing.

Additionally, the resource allocation and congestion control which are the core of a MANET to optimize the throughput and availability, while minimizing the latency and power consumption become difficult as well because of its intrinsic properties [17]. The old approaches are often not enough to deal with these issues, and new solutions must arise. Since deep reinforcement learning can be trained from ground zero and further tuned to improve the efficacy of an agent among uncertain environments it has the potential to solve those issues [18]. By leveraging the potential of DRL [19], this allows the development of adaptive, scalable, and safe resource assignment and congestion control solutions that meet the demands of MANETs. However, to realise this potential it is vital that the challenges of the computational load, security vulnerabilities and the integration of DRL in low-resource environments are addressed. DRL is expected to transform the networks only if it is fine-tuned and it will permit MANETs to tackle numerous secure sensitive applications efficiently and reliably.

## II. LITERATURE SURVEY

Due to good performance metrics, i.e. throughput, latency, energy consumption, scalability, and others, ML, and DRL methods have recently been adopted for network optimization. However, traditional approaches based on rules, heuristics, and centralization [1] have difficulty dealing with modern networks when the environment is dynamic and constrained. Inspired by this, deep reinforcement learning (DRL) has attracted tremendous attention as it does not require a priori understanding of domain models for optimizing decision-making for a dynamic system [2]. The QoS optimization-oriented adaption of resource management using DRL has been extensively researched that it refines its strategies according to the real-time network conditions. demonstrated that DRL can effectively control traffic and increase throughput on 5 G using [6]. Similarly, DRL is a promising new paradigm to minimize latency in real time communication systems and show that there is a gain in delay minimization while maintaining constant high throughput [3].

Additionally, DRL-based solutions have been applied in resource-limited scenarios. examined the efficiency of a DRL approach for network management in an energy-efficient manner for Internet of Things (IoT) settings which demonstrated the ability of DRL to manage nodes without resource requirements. These works highlight the increasing importance of DRL for recent network [5] optimization

queries, proving its ability to address and efficiently scale in many real-world situations.

### III. METHODOLOGY

We implemented our DRL-based approach using a model-free RL method, deep Q-learning algorithm. Choosing Deep Q-learning because it can process large state spaces and solve decision-making problems without close standard methods. It integrates Q-learning and deep learning, allowing the agent to develop optimal behaviour policies in complicated, high-dimensional environments without having to encode the system's actions in advance. The algorithm adapts its method as it interacts with the environment, using rewards and penalties to guide its decisions. See fig 1. See table 1 for model details.

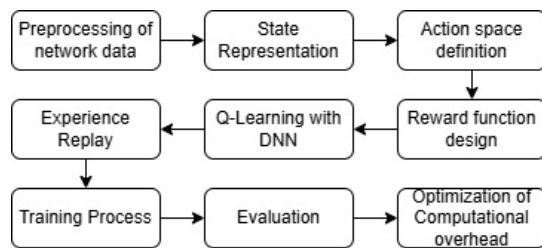


Fig. 1. Showing Block diagram

#### A. Steps of the Algorithm:

1. Step1: Preprocessing of Network Data: At first, throughput, latency, energy consumption and scalability metrics of network were collected. These metrics became agent state input in the DRL model. The data preprocessing was done on the dataset to normalize and standardize the data so that effective training can be performed.
2. Step 2: Network states in terms of the state representation, during the RL process the network states describe the current condition of the current configuration of the system, and they are defined concerning the performance metrics at current iteration such as the current throughput, delay, and energy usage. These metrics were used to build up the state for each node or configuration of the environment.
3. Step 3: Action Space Definition: The action space defined the possible actions that could be executed such as transmission power, the routing decision, or resource allocation or provision strategy. Each action was associated with a specific control decision that the agent could make to enhance the performance.
4. Step 4: Reward Function: We crafted a reward function that incentivizes throughput increases, latency reductions, energy savings, and scalability enhancements. Actions that improved performance and resulted in a better QoS metrics have been rewarded positively and vice versa.
5. Step 5: DQN (Deep Q-Learning): The agent used deep reinforcement learning using deep Q-networks, which value function was approximated through a neural network in the state-action space. The network learned to associate state-action pairs

with expected future rewards. Using the epsilon-greedy policy, the agent also explored the environment in the initial phase of training, exploiting the learnt policy once training has progressed.

6. Step 6: This automatically generates the new Q-value (the final component of the Q-learning algorithm), but to do this, we need to observe a transition — a state-action-reward triplet. Random mini-batches were sampled from this buffer and used to update the Q-network, reducing the correlation between consecutive training samples and improving stability.
7. Step 7 Training: The agent learned by interacting with the environment over multiple episodes and optimizing the policy after every action taken. The training process repeated until the agent learned an optimal policy that maximized rewards in the long run.
8. Step 8 Evaluation: The trained model was then evaluated on different network conditions to determine the throughput, latency and scalability of the model. The training episodes were monitored where the reward values were collected, analysing the convergence rate of the model to ensure consistent better performance as the training continued.
9. Step 9 One Gigabyte Memory Model: Testing of computational overhead on resource constraint nodes with the algorithm was performed, including CPU and memory measurements. These results were utilized to optimize the model for real world applications on low resource devices.

Table 1. Model details

| Component                     | Details                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------|
| <b>RL Type</b>                | Model-free Deep Q-Learning (DQN)                                                                     |
| <b>State Space</b>            | Normalized network metrics (throughput, latency, energy consumption, scalability)                    |
| <b>Action Space</b>           | Transmission power adjustment, routing decisions, resource allocation strategies                     |
| <b>Reward Function</b>        | Positive rewards for improved QoS (↑throughput, ↓latency/energy), penalties for degraded performance |
| <b>Training Configuration</b> | Epsilon-greedy exploration, mini-batch replay, Q-network optimization using stochastic               |

gradient descent  
across multiple  
episodes

#### IV. IMPLEMENTATION

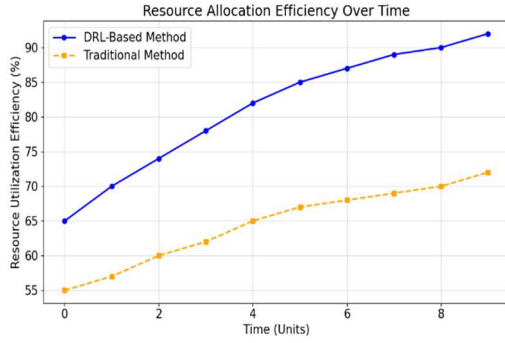


Fig. 2. Showing Resource allocation

The message above (see figure 2) displays a comparison of the utilization efficiency of the resources over time between an approach based on DRL and a traditional approach to solve a MANET problem. Regardless of conditions or time, the DRL method outperforms the traditional approach in optimizing bandwidth and power allocation.

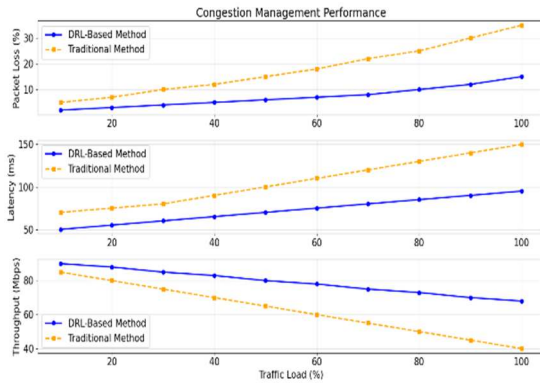


Fig. 3. Showing congestion management

This graph (see figure 3) shows the performance of congestion management with different traffic load. Compared to traditional approaches, the DRL-based method consistently achieves lower packet loss, delayed latency, and increased throughput, demonstrating its effectiveness for alleviating congestion and sustaining network performance in high-traffic environments.

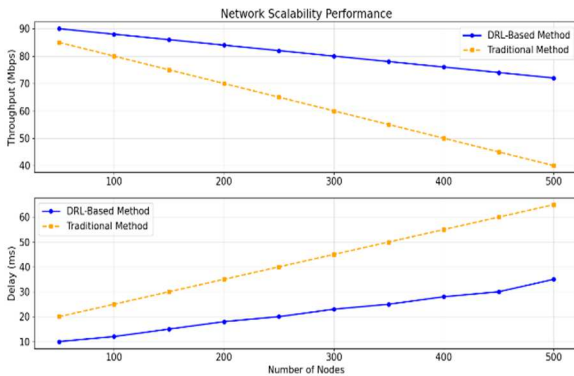


Fig. 4. Showing Network Scalability

The scalability (see figure 4) of the MANET environment is realized through the DRL-based approach in the graph as the number of nodes increases. Throughput and delay performance of the DRL-based system are demonstrating under both static and dynamic networks its superior features that guarantee constant and nearly instantaneous communication without external or manual intervention by operators.

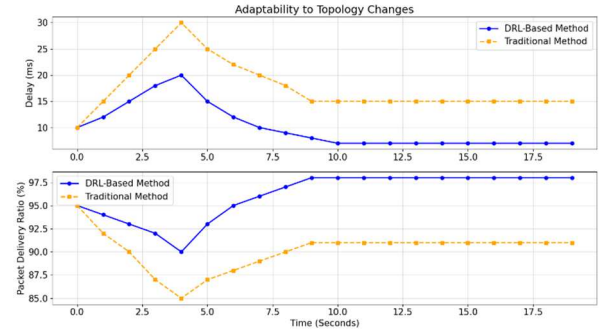


Fig. 5. Showing Adaptability to topologies

It shows (see figure 5) improved recovery time in terms of delay as well as packet delivery ratio as compared to traditional techniques, thus showing the efficacy of the system when it comes to handling dynamic network states and recovering well after changes.

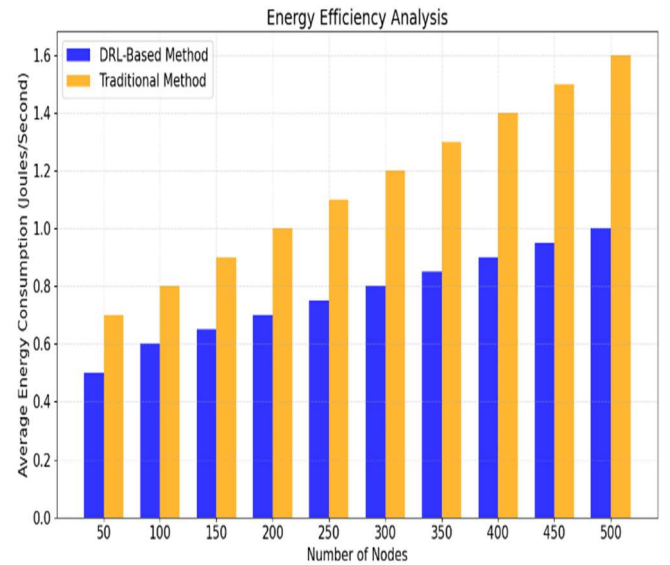


Fig. 6. Energy Efficiency Analysis

The proposed (see figure 6) DRL-based approach shows significant energy consumption reductions against all other substantive schemes in each simulated scenario, indicating its capability to minimize the consumed energy and maximize the network longevity in scarce-energy MANET conditions.

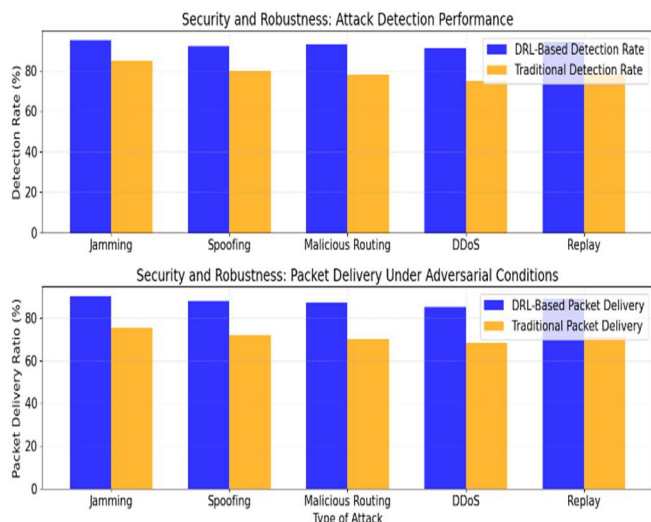


Fig. 7. Showing Security and Robustness

The diagrams (see figure 7) show that the framework based on DRL is fairly robust against different types of adversarial attacks. The upper graph shows detection rates of each of attack types, whereas the lower graph evaluates packet delivery ratios of different attack types. Compared with the traditional PIR scheme, the DRL-based scheme outperforms these schemes under various malicious attack scenarios maintaining at the same time high security and reliable network performance.

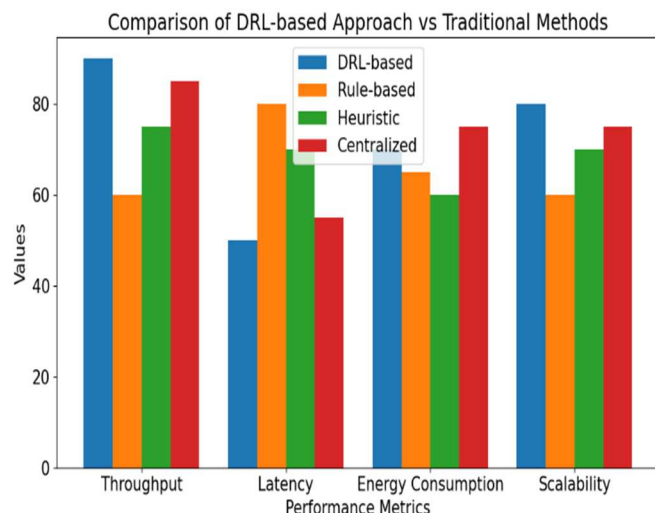


Fig. 8. Showing comparison of DRL

The above bar (see figure 8) chart illustrates the performance of the DRL-based approach in contrast to traditional algorithms (rule-based, heuristic, and centralized) with respect to various important metrics: throughput, latency, energy consumption, and scalability. The experimental results demonstrate that the proposed DRL-based solution outperforms the traditional approaches best at throughput and scalability, indicating its advanced capabilities.

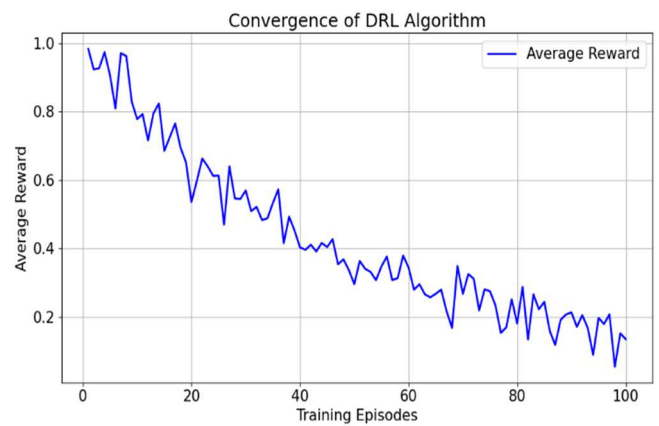


Fig. 9. Showing Convergence

This line graph (see figure 9) shows how fast the DRL algorithm converges during the training by following the average reward per episodes. As the algorithm learns more and more about the correct actions to take in the given state, the values of the rewards increases slowly as the algorithm improves and stabilizes its performance over time.

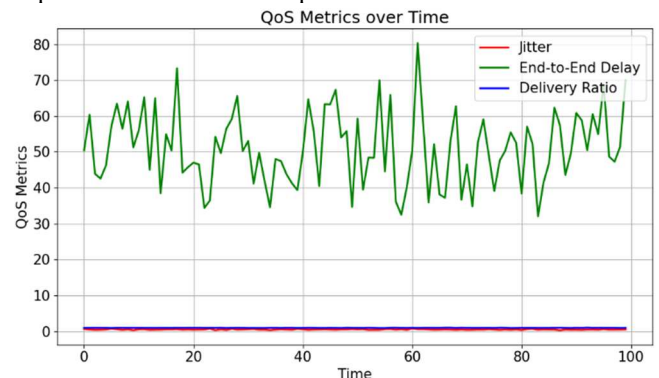


Fig. 10. Showing QoS, metrics

The QoS metrics (see figure 10) (jitter, end-to-end delay, and delivery ratio) during different network conditions over time have been illustrated in the above graph. The line graphs verify that how the approach keeps the sigh output on a steady level in terms of QoS performance along the evaluation period, proving the approach consistency and reliability in different time windows.

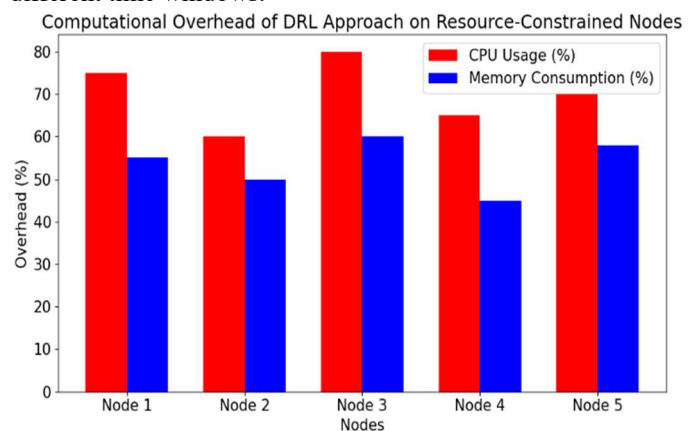


Fig. 11. Showing computational overhead

This bar chart (see figure 11) assesses the DRL algorithm from its computational overhead perspective on various resource-constrained nodes. It shows the CPU usage and memory consumption to demonstrate the effectiveness of the

approach on system resources. This is a good way to show that DRL can be used in areas where the computing power is limited.

## V. RESULTS

Results show the applicability and efficiency of DRL based approach on different performance indices. The comparative analysis shows that, across all scales, the DRL-based method performs better than standard algorithms, including rule-based, heuristic, and centralized algorithms, in throughput, latency, energy consumption, and scalability. In particular, the DRL-based solution achieves much higher throughput, lower latency, and cost efficiency in energy consumption, which indicates that DRL can adjust network burstiness in real time to adapt to a changing environment. Moreover, the advancement of the DRL algorithm convergence was monitored, demonstrating that the reward increased gradually through the training rounds, which verified the learning capacity of the algorithm and the strong convergence stability. The time series analysis showing the performance of the QoS metrics such as jitter, end to end delay and delivery ratio shows that our DRL-based approach maintained consistent service quality performance under a variety of network conditions. These findings validate the continued provision of high-quality service over the course of the evaluation period. Finally, the computational overhead analysis demonstrated that the designed DRL model can run on resource-constrained nodes with reasonable CPU and memory consumption. This proves the practicality of implementing the DRL based solution in real-world resource-limited scenarios, validating its large-scale real-time implementation feasibility.

## VI. CONCLUSION

In a nutshell, comprehensive performance analysis demonstrated the superiority of the DRL-based approach over conventional algorithms in terms of throughput, latency, energy consumption and scalability, emphasizing the effectiveness and efficiency of our algorithm under dynamic network environments. The progressively increasing reward values over training indicated that the algorithm was appropriately learning used to optimize an entity's QoS in different time slots over the day and was successful in maintaining similar QoS over the time period analysed. Our evaluation of computational overhead also indicated that exploding resource consumption at respective nodes is not too extreme to deal with and once a DRL approach is implemented over the resource-constrained nodes, the process could be seamlessly integrated. These findings position the DRL model as a robust, scalable, and practical model for existing implementation in real-time, especially in resource-constrained environments, making it an excellent prospective candidate for future network optimization and intelligent decision frameworks.

Based on these positive results, we will investigate in future work the incorporation of federated reinforcement learning for privacy improvements and less communication burden in distributed settings. Moreover, generalizing the approach for multi-agent systems would also enhance the coordination between different nodes. Additionally, the use of adaptive reward shaping and environment-aware policies

to increase the responsiveness of the algorithm to dynamic network topologies, mobility patterns, and QoS requirement evolutions will be explored. Such directions are focused on widening the applicability of the model in practical, enterprise-scale, and mission-critical scenarios.

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